Stress and Adaptogens

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Abstract

Stress is appropriately defined as a "nonspecific response of a body to any demand". Stress may be classified based on various factors including the type of stressor, duration of the stress response, and also on the type of influence on an individual. Upon exposure to stress and based on the type of stress, the body initiates an adaptive response that is governed by various pathological and genomic pathways. Furthermore, if exposure to stressors is prolonged for years in case of extreme stress, it may lead to a pathological state leading to chronic diseases such as cognitive deficit, accelerated aging, hypertension, and other cardiovascular diseases. Such incidences may require interventions such as nutraceuticals and medicines. Adaptogens are plant extracts that help the body adapt or adjust to chronic exposure to stress. They protect against stress-induced pathophysiological symptoms, accelerate mental functioning, and regulate homeostasis. In this chapter, we provide a basic overview of stress and how adaptogens play an excruciating role in its management. The role of selected adaptogens such as *Rhodiola imbricata, Hippophae rhamnoides, Ganoderma lucidum,* and *Cordyceps sinensis* eliciting various pharmacological actions by acting on different mechanistic pathways is described.

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Stress · Classification of stress · Stress management · Adaptogens

1.1 Introduction

Hans Selye defined stress as nonspecific exertion on the body disturbing body functions and adaptive response involving certain hormones, stress hormones, and defined it as general adaptation syndrome (Selye 1956). The general adaptation syndrome involves alarm reaction sensing stimulus and disturbing body homeostasis, resistance, adaptive response to the stressor and exhaustion, and adaptive failures upon prolonged exposure (Tost et al. 2015). Stress is often nonspecific and complex process and adaptive response is individual-specific. The active process of adaptation against stressors is known as 'allostasis' which involves the preservation of homeostasis via the nonlinear intervention of the autonomic system, metabolic system along with immune system (McEwen and Wingfield 2003). The cumulative effects of multiple stimuli and the dysregulation of the nonlinear mediation of allostasis lead to 'allostatic load' which is responsible for the 'wear and tear' of the brain and body causing damage to organs, tissues, and cellular functioning such as mitochondrial dysfunction and telomere shortening that is associated with accelerated aging.

A stimulus, a stressor, in a biological system initiates cellular events that result in cellular stress. Stressors can be of various types including chemicals, environmental conditions, or biological agents that may activate the hypothalamic-pituitary-adrenal (HPA) axis stress response (McEwen 1998). Cellular stress initiates response including stress hormone secretion, proliferation, and immune response to maintain the well-being state of body and mind leading to adapt the individual to the stressor (Cortez 1991; Cortez et al. 2008; Cortez and Silva 2020). Exposure to stress causes increased levels of hormonal mediators including cortisol and catecholamines such as epinephrine and norepinephrine that are regulated through the hypothalamic pituitary adrenal (HPA) axis along with the sympathetic nervous system (Ajala 2017) (Fig. 1.1).

Stressors impact mood, state of well-being, psychological distress, and health. In healthy individuals, acute stress may be adaptive; however, long-term stress in unhealthy individuals may cause illness (Schneiderman et al. 2005).

The stressors are multi-mechanistic in altering gene expression. They are responsible in the activation of epigenetic mechanisms and gene transcription through glucocorticoids, along with modifications of histones as well as methylation and hydromethylation of CpG residues in DNA, thereby repressing and activating genetic factors such as retrotransposons (McEwen et al. 2015). The primary target of stressful environment is the brain, and stimulation of glucocorticoids and excitatory amino acid neurotransmitters through stressors leads to alteration of the neuronal architecture by affecting the dendritic growth, synapse density, as well as inhibition of neurogenesis (Lupien et al. 2018). The three major regions of brain that

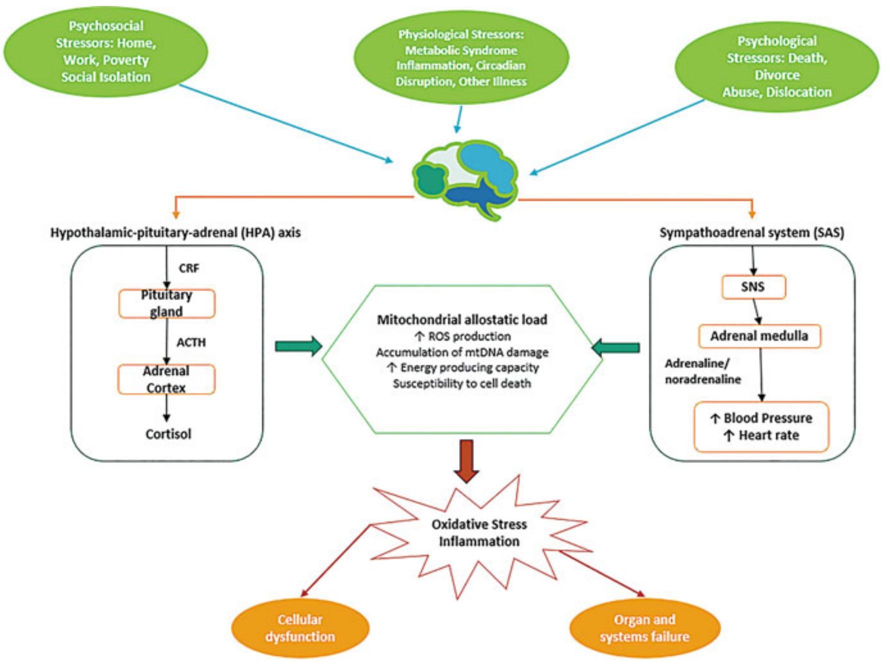

Fig. 1.1 Mechanism of chronic stress response system

play an important role in the regulation of the HPA stress response include hippocampus, amygdala, and prefrontal cortex (McEwen et al. 2015). The presence of glucocorticoid and mineralocorticoid receptors in the hippocampus evidently illustrates the effect of stress through glucocorticoids by shrinking the dendritic growth and spinal loss along with inhibition of neurogenesis in the dentate gyrus region (McEwen et al. 2016). In the amygdala, the effect is based on the type of stress such as increased spine density on basolateral neurons is observed during acute traumatic stress and chronic stress leads to dendritic extension and loss of spines. The effects of stress on the prefrontal cortex provide an insight to the memory loss associated with age as chronic stress leads to dendritic shrinkage and debranching of neurons (Arnsten 2009).

It is well-documented that stress stimulus is present in physiological and pathological processes (Prado et al. 2006) and is often mistakenly defined as illness. In actuality, stress is well defined as a stimulus inducing biological alarm and prolonged exposures may lead to a pathological state, i.e., illness or diseases. Moreover, anxiety and stress are defined as the same, but stress is a response to stimulus and may lead to anxiety (Silva et al. 2017; Rom and Reznick 2016; Sevillano-García et al. 2007; Cortez 1991).

The need for the management of chronic stress is required when long-term exposure could lead to a various of stress-related diseases such as hypertension, anxiety, depression, cognitive dysfunction, and chronic fatigue syndrome (Ajala

2017). The individual response to the same type of stress may be varied due to complex responses exhibited by the cells and tissues and genetic factors. Therefore, the management of stress cannot be straightforward or universal. Every day, organism deals with stress in one or the other way. The first step for effective management of stress is recognizing stress and the source of stress and identifying whether it's acute or chronic stress. If stress leads to uneasiness in life and disturbances in physiological or psychological functions, identifying the stage of stress would be beneficial in stress management.

The management of acute stress, though not easy, can be regulated through relaxation, meditation, yoga, avoiding stressful environments, etc. However, in case of long-term adaptation, pharmacological interventions may be required. Numerous herbs are well documented to alleviate stress and extreme stress including *Cordyceps sinensis*, *Ganoderma lucidum*, *Ginkgo biloba*, *Ginseng*, *Hippophae rhamnoides*, *Rhodiola imbricata*, etc.

In this chapter, types of stress and the management of stress through adaptogenic plants and medicinal mushrooms have been elaborated with special reference to extreme stress environments, i.e., high altitude. These adaptogenic plants synthesize a class of phyto-compounds that help in enhancing the body's resistance to various stressors including emotional, environmental, and physical stress.

1.2 Classification of Stress

Stress can be classified according to the nature of the stressor, whereby the stressor can be physiological or psychological (Kogler et al. 2015). Physiological stress is induced by an unpleasant sensory or an emotional experience that is usually associated with potential bodily harm or damage to body physiology, for example, pain, hunger, oxidative stress, etc. (Zubieta and Stohler 2009). On the contrary, psychosocial stress is triggered by social situations of threat such as social assessment, rejection, and even by incidents that highlight goal-oriented accomplishments (Pruessner et al. 2010) Table 1.1.

Another classification of stress is based on the type of influence a stressor has that is reflected on an individual. Good stress or stress that can be described as a positive response to a stressor is known as eustress. Simultaneously, a negative response to a stressor is known as distress, or bad stress (Gong and Geertshuis 2023). The type of response to distress is severely high and may lead to psychological or physical damage (Table 1.1). Distress can also be further classified into acute stress and chronic stress, according to the duration of the stress response. Acute stress is when the stress response is high in intensity, but remains short-term, whereas, chronic stress is when the stress response may not be highly intense, but lasts for a prolonged interval of time (Lu et al. 2021) Table 1.1.

Table 1.1 Classification of stress

Classification of Stress	
A. Based on the nature of the stressor	
Physiological stress	Psychological stress
B. Based on the influence of stress on an individual	
Eustress	Distress
C. Based on the duration of stress response	
Acute stress (short-term)	Chronic stress (long-term)

1.3 What Are Adaptogens?

The original definition of adaptogens stated that these are agents that "increase resistance to a broad spectrum of harmful factors (stressors) of different physical, chemical, and biological natures" (Brekhman and Dardymov 1969). These are substances that improve the state of nonspecific tolerance to stress (Wagner et al. 1994). Drugs are classified as adaptogens if they exhibit anti-stress effects including anti-fatigue, antidepressant, and anti-infectious; stimulating effects that are responsible for increasing work and mental performance; however, they do not affect the normal functioning of the anabolic system and exhibit no adverse effect such as withdrawal syndrome (Brekhman and Dardymov 1969).

Adaptogens are known to have complex chemical structures of phenolic or tetracyclic triterpenoids/ steroids. Some of the phenolic adaptogens such as rhodioloside, rosavin, syringin, triandrin, and tyrosol are derivatives of phenylpropanoids and phenylethane. Since adaptogens are responsible to mediate the stress response system, they structurally mimic catecholamines that initiate the regulation of sympathoadrenal system (SAS) (Panossian and Wikman 2010). In addition, compounds such as ginsenosides, eleutheroside, cucurbitacin R diglucoside, sitoindosides, etc. are tetracyclic triterpenoids that chemically resemble the structure of corticosteroids, also known as the stress hormone (Munck et al. 1984). Furthermore, extracts from *Rhodiola* sp., such as rosiridin, are monoterpene glucosides that act as inhibitors of monoamine oxidases A and B (Van Diermen et al. 2009).

1.4 Possible Mechanism of Adaptogens in Stress Management

Adaptogens, as the name portrays, are regulators of metabolic pathways that increase an organism's potential to adapt to any environmental or any other external factors. According to the available literature, several mediators such as stress-activated protein kinases, FOXO transcription factor, heat shock proteins (HSP), and nitric oxide play an important role in the regulation of apoptogenic potential of various plant extracts (Alexander Panossian 2017).

The plant extracts provide a safe treatment as they stimulate the release of HSP70 without being cytotoxic in nature. Inhibition of HSF1 and HSP70 expression has been associated with age-related diseases and neurodegenerative disorders, including Alzheimer's disease, and other metabolic diseases such as diabetes (Asea et al. 2013). Adaptogens also act as inhibitors of stress-activated kinases and have been reported to significantly decrease the levels of phosphorylated forms of SAPK/JNK (Wiegant et al. 2009). In addition to this, adaptogens are also observed to be responsible for the inducing the translocation of FOXO (DAF-16) into the nucleus, thereby favoring a transcriptome that is longevity-biased and stress-resistant (Perez et al. 2012).

1.5 Role of Adaptogens: Few Examples

The role of adaptogens from medicinal plants and mushrooms such as *Rhodiola imbricata*, *Hippophae rhamnoides*, *Ganoderma lucidum*, and *Cordyceps sinensis* has been discussed below and their probable mechanism has been specified in Table 1.2.

1.5.1 Rhodiola imbricata

Rhodiola imbricata grows at high altitude and is from the family Crassulaceae and subfamily Sedoideae which is widely distributed throughout Europe and Asia**.** Komarov (1939) has defined 20 species and it is estimated that there are about 200 species of Rhodiola worldwide. Rhodiola rhizome is widely used in China, Soviet Republics, Tibet, and Mongolia to alleviate stress, extreme high-altitude stress, longevity, and endurance (Kelly 2001). Further, it has been reported to prevent fatigue and gastrointestinal illness (Kelly 2001; Petkov et al. 1986). *Rhodiola species* have been extensively investigated in animal and human studies indicating adaptogenic and anti-stress properties and found to be effective against lifestyle disorders and diseases (Panossian et al. 2007; Gupta et al. 2008, 2009, 2010; Ajala 2017; Li et al. 2017a, b; Shieh et al. 2018; Zheng et al. 2019; Agapouda et al. 2022; Ivanova Stojcheva and Quintela 2022). The medicinal properties are attributed to *Rhodiola rosea* owing to the presence of certain bioactive substances like rosavins (rosavin, rosarian, and rosin), phenylethanolderivatives (salidroside, tyrosol), organic acids, flavonoids, tannins, and phenolic glycosides (Khanum et al. 2005; Mao et al. 2007). Its protective efficacy against mental and physical stress, toxins, and cold in experimental animals and humans is well documented (Khanum et al. 2005; Saratikov and Krasnov 1987; Kurkin and Zapesochnaya 1986; Petkov et al. 1986, 1990; Saratikov et al. 1978). Rhodiola acts as an antidepressant and psychostimulant (Saratikov and Krasnov 1987; Konstantinos and Heun 2020) and is useful in alleviating neurological disorders (Table 1.2).

Table 1.2 Mechanism of action of few adaptogens along with their intended pharmacological use

Name of Adaptogen	Mechanism of Action	Intentional use	References
Rhodiola imbricata	• Decreases production of ROS • Inhibits H_2O_2 • Stimulates interleukin-6 (IL-6) and tumor necrosis factor (TNFα) • Increases production of NO • Induces apoptosis and cell death	o Anti-cytotoxic o Hepatoprotective o Antioxidant o Anti-stress	(Zhuang et al. 2019) (Bhardwaj et al. 2018)
Hippophae rhamnoides	• Inhibits NMDA receptors • Antagonises Serotonin (5HT) receptor • Reduces corticosterone levels • Reduces the levels of dopamine and norepinephrine • Reduces oxidative stress • Upregulates endogenous antioxidants	o Anti-stress o Antidepressant o Antioxidant and immunomodulatory o Cardioprotective o Antibacterial and antiviral	(Suryakumar and Gupta 2011) (Bae et al. 2016)
Ganoderma lucidum	• Enhances levels of neurotransmitters (catecholamines, serotonin, glutamate) • Prevents increase in glucocorticoid and α-synuclein levels • Improves neuroplasticity by upregulating CREB/p-CREB/BDNF expression via ERK1/ERK2 induction • Attenuates pro-inflammatory cytokines (NFκB, IL6, TNFα) • Maintains oxidative status by normalizing antioxidant enzymes (MDA, GSH, SOD)	o Neuroprotective o Antioxidant, anti-inflammatory, and immunomodulatory o Anti-stress	(Sharma et al. 2019) (Sharma and Tulsawani 2020)
Cordyceps sinensis	• Improves the activity of anti-oxidative enzymes such as SOD, catalase, GSH-Px and lowers the activity of lipid peroxidation and monoamine oxidase B • Increases production of pro-inflammatory cytokines from activated microphages • Regulation of the *foxo* signaling pathway (hsa04068) and HIF-1 signaling pathway (hsa04066)	o Antidepression o Anti-oxidative stress o Antiaging/ senescence o Anticancer o Anti-fatigue/ asthenia	(Chen et al. 2013) (Jordan et al. 2008) (Zhang et al. 2022)

1.5.2 Hippophae rhamnoides

Hippophae rhamnoides, commonly known as seabuckthorn, belongs to family Elaeagnaceae and genus *Hippophae*. It is a wild thorny deciduous shrub, grows at high altitude (2500–4000 m) in adverse climatic conditions, and is native to Europe and Asia (Rousi 1970). In India, it grows naturally in Ladakh, Uttarkhand, Himachal Pradesh, Jammu and Kashmir, Sikkim, and Arunachal Pradesh regions (Dhyani et al. 2010). There are six species of seabuckthorn found in the world, of which three species are reported in India, viz., *Hippophae rhamnoides subspecies Turkestania Rousi*, *Hippophae salicifolia*, and *Hippophae tibitana*. Among these species, *H. rhamnoides* has been extensively studied and many of the reports are from Russia and China (Li and Beveridge 2003). *Hippophae rhamnoides* has been reported to possess adaptogenic potential and initiate adaptive response to stress (Saggu et al. 2007; Tulsawani et al. 2010; Patel et al. 2012). It is reported to possess antioxidant and anti-stress activity (Suryakumar et al. 2002; Narayanan et al. 2005; Liao et al. 2018; Ji et al. 2020). The medicinal properties of extracts of *Hippophae rhamnoides* leaves are attributed due to the presence of high content of flavonoids, tannins, and triterpenes (Kallio et al. 2002). The major flavonoids found in the leaves of seabuckthorn include quercetin, kaempferol, and isorhamentin, catechin, and rutin (Zu et al. 2006). It is documented that *Hippophae rhamnoides* L. possesses neuroprotective activity and is effective in aging and neurodegenerative diseases including Alzheimer's disease (Xu et al. 2005; Shivapriya et al. 2015; Bala et al. 2017; Cho et al. 2017; Dong et al. 2020; Ladol and Sharma 2021; Veronique et al. 2022; Wang et al. 2022) (Table 1.2).

1.5.3 Ganoderma lucidum

Mushrooms are known to possess antioxidant property; one such mushroom is *Ganoderma lucidum,* commonly found at high altitudes. *Ganoderma lucidum,* also referred as lingzhi or reishi medicinal mushroom, belongs to family ganodermataceae and has been traditionally used in Chinese Medicine and Asian countries for centuries due to its immunomodulatory and adaptogenic properties (Gao et al. 2004; Mizuno et al. 1991; Wang et al. 2002). The pharmacological properties of *Ganoderma lucidum* are assigned due to the presence of array of bioactive compounds including adenosine, coumarin, ergosterols, ganoderic acids, lactones, mannitol, organic germanium, polysaccharides, triterpenoids, and rare minerals (Boh et al. 2007). These phytoconstituents help in improving blood circulation, eliminate fatigue, enhance energy, strengthen the immune system, and help in detoxification and many aliments and lifestyle disorders (Liu et al. 2009, 2017; Min et al. 1998; Sliva 2003; Wihastuti and Heriansyah 2017; Wu et al. 2016; Xu et al. 2017; Zhao et al. 2018). *Ganoderma lucidum* possesses anti-stress (Tulsawani et al. 2014, 2015) and adaptogenic potential (Sharma et al. 2019). *Ganoderma lucidum* prevented extreme hypobaric hypoxia-induced oxidative stress and memory impairment and has been found to be effective in oxidative stress and neurodegenerative diseases including Parkinson's diseases in experimental animals (Zhou et al. 2012; Zhang et al. 2014; Xuan et al. 2015; Sun et al. 2017; Koganti et al. 2018; Ren et al. 2019; Abate et al. 2020; Sharma and Tulsawani 2020; Jatwani and Tulsawani 2021) (Table 1.2).

1.5.4 Cordyceps sinensis

In general, mushrooms have been widely used as alternative remedies to cure various ailments. *Cordyceps sinensis* (*C. sinensis*) (Berk.) Sacc, kingdom Fungi, division

Ascomycota, family Ophiocordycipitaceae, commonly known as caterpillar fungus, has been used as a stimulant for endurance, longevity, and vitality in traditional medicine system (Zhu et al. 1998). It naturally occurs at an altitude of 3600–5000 m in Northern Himalayan regions due to its adaptogenic potential (Li et al. 2011). Phytoconstituents identified in *Cordyceps sinensis* such as cordycepin, polysaccharides, phenols, and flavonoids are known to exhibit pharmacological effects including antioxidant potential and immunomodulatory properties (Wasser and Weis 1999; Li et al. 2001; Kuo et al. 2001). It has been demonstrated that aqueous extract of *C. sinensis* induced dose-dependent tolerance in human lung epithelial cells by reducing oxidative stress via NRF2 induction under low oxygen environment (Singh et al. 2013). It is well documented that *Cordyceps sinensis* possesses anti-stress and adaptogenic activity against severe and extreme stressors (Koh et al. 2003; Sharma and Tulsawani 2022). Cordyceps sinensis has been shown to prevent oxidative stress and neurodegenerative in many experimental conditions (Jin et al. 2004; Li et al. 2012; Olatunji et al. 2016; He et al. 2018, 2019, 2020, 2021; Liu et al. 2022; Wu et al. 2022; Zhang et al. 2022) (Table 1.2).

1.6 Conclusion

Here we presented a basic overview of stress and role of few adaptogens. Stress is a state in which the bodily homeostasis is compromised. The occurrence of stress systematically involves the activation of HPA axis that is triggered in the presence of a stressor. According to the type of stressful stimuli, the conception of stress can be summarized into different types of stress responses such as psychological stress, oxidative stress, metabolic stress, emotional stress, etc. The stress response system is a complex process that revolves around multiple mediators and includes both genomic and nongenomic interacting pathway systems. Acute stress responses are adaptable and beneficial; however, exposure to a chronic stressor can cause bodily harm and lead to several life-threatening diseases in extreme cases. Management of such chronic stress may require external force, and thereby, provides a need for the development of compounds such as adaptogens.

Adaptogens are responsible for improving stress-induced weakness and fatigue, enhancing mental well-being, and reducing pathophysiological impairments caused by stress-induced diseases. The anti-stress potential of adaptogens such as *Rhodiola imbricata, Hippophae rhamnoides, Ganoderma lucidum,* and *Cordyceps sinensis* as mentioned in the chapter is regulated by maintaining homeostasis through multi-mechanistic approach including the mediation of key factors involved in the stress response system, such as glucocorticoids, corticosteroids, nitric oxide, and protein kinases. Besides their anti-stress activity, adaptogens also exhibit beneficial potential against various pharmacological defects including anti-cytotoxic, antidepression, cognitive enhancement, cardioprotective, etc. As the beneficial effects of adaptogens are being investigated, a combination of different plant herbs along with other dietary supplements may provide a potential synergistic effect against the physical and psychological response to chronic stress. Therefore, further investigations are required to fully explore the potential of adaptogens against the management of chronic stress and to provide long-term stress relief.

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